



Learn



Practice

Force and laws of motion: Construction

Contents and concept route



Friction opposes relative sliding at a contact

Subject: Physics

Topic: Laws of Motion

Concept: Friction opposes relative sliding at a contact

Concept difficulty: Medium concept

1

Before giving friction a direction, imagine the contact made perfectly slippery and ask which way the surfaces would slide relative to each other; friction opposes that, and is zero if there is no such tendency.

Medium concept

Conditions for this construction

- Can choose one body and identify external contact interactions acting on it.

Prerequisite reference

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Why this section matters for the question set

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- 1 The step the hard question turns on** Choose the body or system and name the exact contact before deciding friction; for a contact between two bodies, keep the two friction forces as an equal-and-opposite third-law pair on different bodies.

Why valid: Friction belongs to one contact interaction. The force on A by B and the force on B by A are a third-law pair, so neither may be moved onto the other body's free-body diagram.

Result: chosen body/system; named contact and partner; tangent direction; friction owner and receiver fixed.

- 2 Imagine the relative sliding, or tendency to slide, at that named contact if tangential friction were absent; point friction on the chosen body opposite that relative tendency.**

Why valid: Contact friction resists relative sliding between the surfaces.

Result: contact-specific slip tendency -> friction direction; direction need not oppose the body's velocity.

3 Check a conveyor-belt case where the belt can frictionally accelerate an object in the same direction as the object's motion.

Why valid: The counterexample falsifies the shortcut 'friction always points opposite velocity'.

Result: friction direction follows contact slip tendency, not a universal page-direction rule.

4 Decide whether any tangential friction is required at all; allow friction to be zero when the contact has neither relative sliding nor a tendency to slide.

Why valid: Friction is a responsive tangential contact force, not an automatic nonzero force merely because two rough surfaces touch.

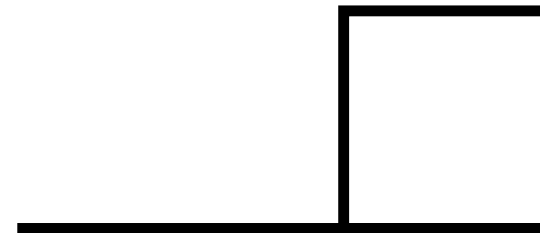
Result: contact-state check -> friction may be zero when no tangential force is required.

Book never slips on the tray

steady speed
 v constant



speeding up
a forward



Show the tray carried at steady speed and then speeding up.

Stage 1 of 3

Show the book on the tray in its two stages so friction's direction is found from the slip tendency, including the zero case.

The key step

Friction is a contact force along the surface that opposes relative sliding, or the tendency for relative sliding, between the contacting surfaces; it is not defined by the object's overall velocity alone.

Formula reference • meaning and conditions



Worked explanation

A book rests on a tray. A waiter first carries the tray at steady speed in a straight line, then speeds up forward; the book never slips. Find the friction force of the tray on the book in each stage, including whether it can be zero.

Follow the operation, its justification and its result.

1. Choose the book as the body; the contact is the tray's top surface, and friction acts along it.
2. Steady carry: if the tray became perfectly slippery, the book would keep moving with the tray, so there is no slip tendency and the friction is zero.
3. Speeding up: without friction the book would keep its old speed while the tray moved ahead, so the book would slip backward relative to the tray.

4. Static friction opposes that tendency, so the tray's friction on the book points forward, the same way the book is moving.

Result

During the steady carry the friction is zero; while the tray speeds up, static friction on the book points forward.

Check

The book speeds up forward and friction is the only horizontal force on it, so by $F_{\text{net}} = m a$ that force must point forward, which agrees.

Mistake clinic • diagnose and repair

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1-2-3 QUICK CHECK

1 • CHECK

CHECK: For a chosen body, name the other surface at the contact and state which way that surface would slip relative to the body if friction vanished.

2 • APPLY

APPLY: A conveyor belt moves right faster than a box on it. Predict the friction on the box and the equal-and-opposite friction on the belt.

3 • CONNECT

CONNECT: Carry this contact ownership into the quantitative test: after direction is fixed, the next question is how much friction the no-slip motion requires and whether the contact can supply it.

Now you do one

A box is placed on a conveyor belt moving to the right faster than the box. Before the box matches the belt speed, what direction is the friction force on the box?

Practice this concept in Core2

INJSO · 2023 · 28 Jan 2023 · Q24

Quantitative static and kinetic friction conditions

Subject: Physics

Topic: Laws of Motion

Concept: Quantitative static and kinetic friction conditions

Concept difficulty: Hard concept

1

Find the actual normal force and the friction that rest would need, compare that need with $\mu_s N$, and use μN as the friction value only at impending slip or once sliding.

Hard concept

Section 2 of 3

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Conditions for this construction

- Can determine friction direction from relative slip tendency.
- Can write Newton-II equations for a chosen body.

Prerequisite reference

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Why this section matters for the question set

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- 1 **Build the chosen-body free-body diagram and solve the perpendicular dynamics for N before using any friction law: do not assume $N = mg$; $N=mg$ only in the special level case with no other perpendicular force or acceleration; a downward added component raises N , an upward component lowers it, and if the required contact force would be negative the contact is lost so $N=0$ and friction=0.**

Why valid: A unilateral contact can push but not pull. Its normal reaction is whatever the perpendicular Newton-II equation requires while contact persists; friction capacity is then tied to that actual N .

Result: perpendicular equation $\rightarrow$ actual $N \geq 0$; level examples $N = mg + F_{\text{down}}$ or $N = mg - F_{\text{up}}$; contact lost $\rightarrow N = 0$ and $f = 0$.

- 2 Assume no relative sliding provisionally, choose a body or a whole connected system deliberately, and solve the tangential Newton-II equation for the friction demand at each rough contact.**

Why valid: Static friction responds to the no-slip dynamics. A system boundary can remove internal contact pairs, but it cannot create friction capacity at a smooth contact or move a rough contact from one body to another.

Result: chosen boundary + no-slip dynamics $\rightarrow$ signed f_{required} at each actual rough contact.

- 3 Compare each $|f_{\text{required}}|$ with $\mu_s N$; on an incline this comparison can be turned into the angle condition for the onset of sliding rather than setting $f = \mu_s N$ from the start.**

Why valid: The static-contact model permits any required value only up to the limiting magnitude.

Result: $|f_{\text{required}}| \leq \mu_s N \rightarrow$ static feasible; equality only at impending slip; after sliding use the kinetic model. Equivalent symbolic form: $|f_{\text{required}}| \leq \mu_s N$.

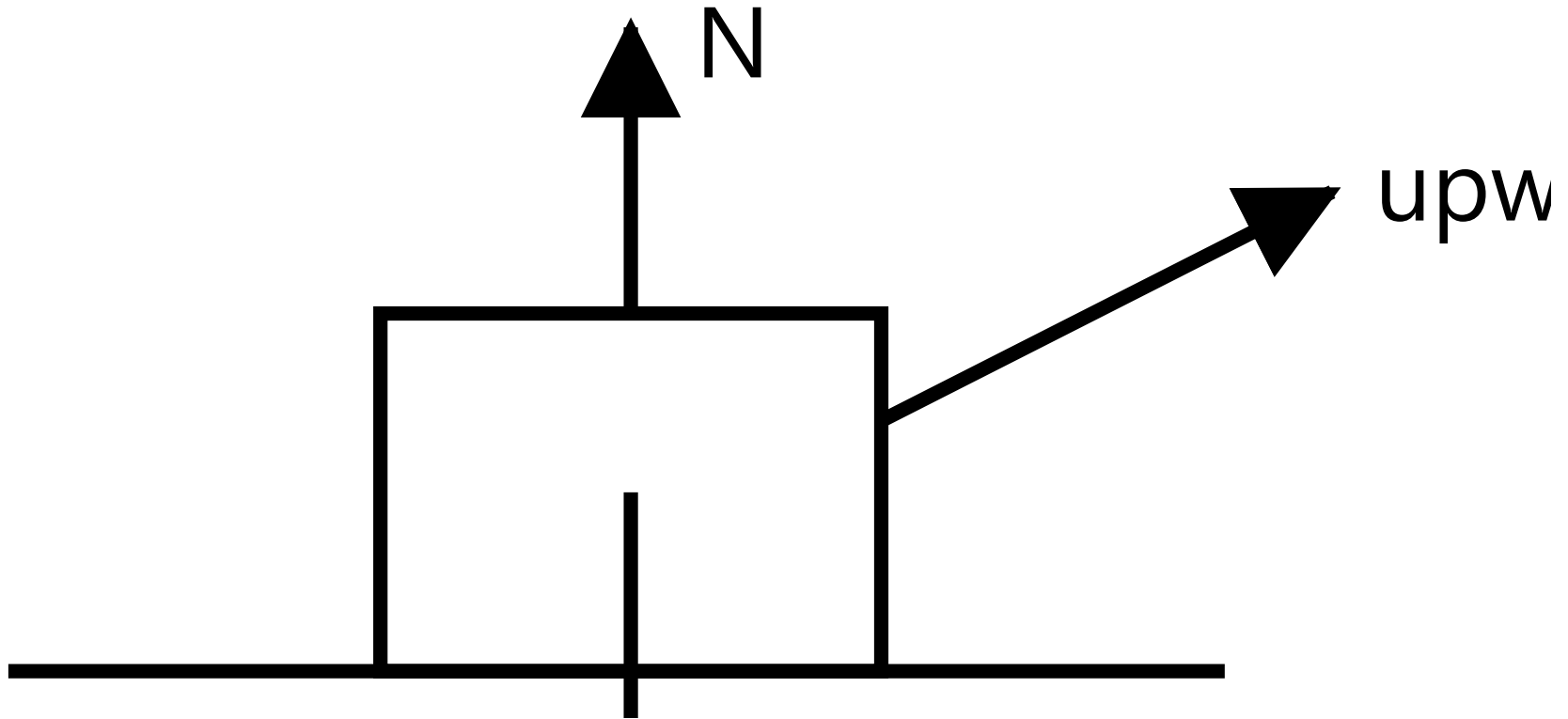
- 4 The step the hard question turns on At the threshold use $|f_s| = \mu_s N$; once relative sliding is established use $|f_k| = \mu_k N$ opposite the sliding, and if several contacts change state, update N and the friction model at each contact before continuing.**

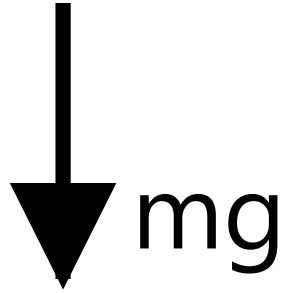
Why valid: Static-below-limit, impending-slip, kinetic-sliding and contact-lost are different contact states. A

multi-contact process may switch which contact is static or sliding, so stale μN values cannot be carried across a transition.

Result: state map: static below limit $\rightarrow f_{\text{required}}$; impending $\rightarrow \mu_s N$; sliding $\rightarrow \mu_k N$; contact lost $\rightarrow N=f=0$.

1 · Solve perpendicular dynamics





Use $\Sigma F_{\perp} = ma_{\perp}$. $N = mg$ only when

If the equation requires $N < 0$: contact

CONTACT LOST $\rightarrow N = 0$ and $f = 0$

Choose the body and solve the perpendicular dynamics before touching a friction formula.

Stage 1 of 4

Build the complete dry-friction decision from actual normal reaction through static demand, impending slip, kinetic sliding and contact loss.

The key step

Static friction is a responsive contact force: solve for the friction required to prevent relative slip and compare it with the limiting value $\mu_s N$. Equality is a threshold condition, while the kinetic model is used only after sliding is established.

Formula reference · meaning and conditions



Worked explanation

JEE (Advanced) · 2020 · Paper 1 · Q13 · Official past paper, faithful restatement of the original

A uniform metre scale is supported horizontally on two index fingers initially at 0.00 cm and 90.00 cm. Both fingers are moved slowly toward the centre. The slipping finger alternates because static and kinetic friction differ. With $\mu_s=0.40$ and $\mu_k=0.32$, find the final distance x_R of the right finger from the 50.00 cm centre at the instant it stops and the left finger starts slipping again.

Follow the operation, its justification and its result.

1. **Treat the slowly moved metre scale as quasistatic and label the two finger supports by their current coordinates and normal reactions N_L and N_R .**

Why valid: At every instant the scale has negligible translational and angular acceleration, but the support reactions change as the finger positions change.

Result: $N_L + N_R = mg$; their ratio is set by support positions.

2. **For each current pair of support positions, obtain N_L and N_R from force and torque balance before writing friction.**

Why valid: Friction capacities are μN , so a fixed normal reaction while a support moves would make the threshold condition inconsistent with the changing support geometry.

Result: With the right finger at 90 cm, $N_L/N_R = 40/(50-l)$; the first switch gives $l = 18$ cm.

3. **After the first switch, hold the left finger at 18 cm while the right finger slips inward and recompute the support ratio $N_R/N_L = 32/(r-50)$.**

Why valid: The support geometry has changed, so the normal reactions—and therefore both friction capacities—must be recalculated for the second stage rather than carried over.

Result: $N_R/N_L = 32/(r-50)$.

4. **At the second stick-slip transition use kinetic friction at the slipping right contact and limiting static friction at the sticking left contact: $\mu_k N_R = \mu_s N_L$; this gives $r-50 = 25.6$ cm and swaps which finger**

slips.

Why valid: The friction law is contact-state specific: a sliding contact uses μ_{kN} while a sticking contact can rise only to μ_{sN} . Equality marks the instant the roles switch.

Result: $x_R = 25.6$ cm.

Result

25.6 cm.

Check

The result is less than the initial 40 cm offset of the right finger and greater than zero, consistent with inward motion.

Mistake clinic • diagnose and repair



1-2-3 QUICK CHECK

1 • CHECK

CHECK: State the four contact states in order and the friction rule for each: static below its limit, impending slip, established sliding, and contact lost.

2 • APPLY

APPLY: A level block has weight 50 N and is pulled 20 N upward as well as 12 N sideways. With $\mu_s=0.5$, find N, the static limit, and the actual friction if it stays at rest.

3 • CONNECT

CONNECT: In a two-contact problem, after one contact starts sliding, recompute the normal reactions and identify which contact is now static before writing the next μN equation.

Now you do one

A contact has known normal reaction N and coefficient μ_s . The no-slip force equation requires tangential friction f_{required} . State the test for whether static contact is possible, when equality with $\mu_s N$ is justified, and what changes once sliding is established.

Practice this concept in Core2

IIT-JEE · 2007 · Paper I · Q10

IIT-JEE · 2008 · Paper II · Q33

IIT-JEE · 2011 · Paper I · Q41

JEE (Advanced) · 2014 · Paper 2 · Q19

JEE (Advanced) · 2020 · Paper 1 · Q13

JEE Main · 2026 · Session 2 · 06 Apr 2026 · Shift 2 · Q46

INJSO · 2011 · 29 Jan 2011 · Q22

A bounded angular factor and an independent total contribution

Subject: Physics

Topic: Laws of Motion

Concept: A bounded angular factor and an independent total contribution

Concept difficulty: Medium concept

1 A partial factor cannot fix a total that also depends on an independent input.

Medium concept

Section 3 of 3

← Previous

Conditions for this construction

- The formula and arbitrary units are stipulated.
- c is an independent contribution; θ is held fixed.
- No real force, energy barrier, or friction prediction follows.

1 Hold θ at 60 degrees.

Why valid: The requested comparison changes c alone.

Result: $g = \cos(60 \text{ degrees}) = 1/2$ in both cases.

2 Evaluate the explicitly given partial term.

Why valid: Squaring is a chosen definition in this toy, not a universal physical law.

Result: $-g^2 = -1/4$.

3 Add each independent contribution.

Why valid: The total is defined as this partial term plus c.

Result: At $c=0$, $E = -1/4$; at $c=1$, $E = 3/4$.

4 Compare the outputs and the claim.

Why valid: Two different totals with identical g refute uniqueness from g alone.

Result: The same angular factor does not determine total E; no physical barrier was calculated.

theta $\rightarrow$ given factor $g = \cos$
independent contribution c

Distinguish the angle input from the independent contribution.

Stage 1 of 3

Show the two independent inputs to a stipulated toy total.

Keep alignment separate from total energy

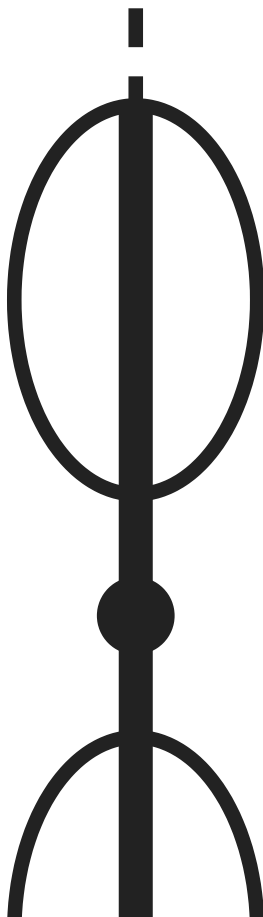
Look along the fixed reference axis. The two centres project to the same origin. The solid test direction turns relative to the dashed reference direction. This is a direction diagram, not a measured field or coupling integral.

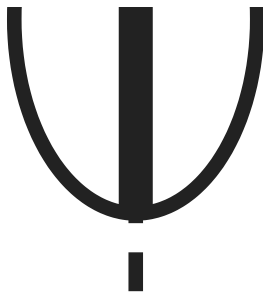
Relative angle θ (degrees)



Independent toy energy contribution c







Axis points towards/away from viewer.

$\theta = 0^\circ$; given factor = 1; toy $E = -1$.

Declared demonstration only: $g = \cos(\theta)$; $E = -g^2 + c$ in arbitrary toy units. The square is an explicitly chosen example, not a universal law. Hold θ fixed and vary c : the same g permits different E . Predict this before moving c . No physical barrier is computed.

The key step

A partial factor cannot fix a total that also depends on an independent input.

Worked explanation

Given the dimensionless demonstration $g = \cos(\theta)$, $E = -g^2 + c$, take $\theta = 60$ degrees. Compare $c = 0$ and $c = 1$. Does the same g determine a unique E ? This is a chosen toy expression, not a physical energy law or barrier calculation.

Follow the operation, its justification and its result.

1. Hold θ at 60 degrees.

Why valid: The requested comparison changes c alone.

Result: $g = \cos(60 \text{ degrees}) = 1/2$ in both cases.

2. Evaluate the explicitly given partial term.

Why valid: Squaring is a chosen definition in this toy, not a universal physical law.

Result: $-g^2 = -1/4$.

3. Add each independent contribution.

Why valid: The total is defined as this partial term plus c .

Result: At $c = 0$, $E = -1/4$; at $c = 1$, $E = 3/4$.

4. Compare the outputs and the claim.

Why valid: Two different totals with identical g refute uniqueness from g alone.

Result: The same angular factor does not determine total E; no physical barrier was calculated.

Result

At $\theta=60$ degrees, both have $g=1/2$; E changes from $-1/4$ to $3/4$ as c changes from 0 to 1.

Check

At $\theta=90$ degrees, $g=0$ and $E=c$, so c still changes the total without changing g.

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1-2-3 QUICK CHECK

1 • CHECK

At $\theta=90$ degrees, $g=0$ and $E=c$, so c still changes the total without changing g.

2 • APPLY

At $\theta=60$ degrees and $c=-1$, the toy gives $E=-5/4$.

3 • CONNECT

Identify both independent inputs before treating a partial factor as a total.

Question-specific diagnosis and repair



Now you do one

Hold theta at 90 degrees and compare $c = -1$ with $c = 1$. Can g alone identify the total?